

TABLE 1
COMPARISON OF INDUSTRY-READY COMMUNITY BOARDS

Board Name/ Part Number	Family Name	Board Manu- facturer	Silicon Manu- facturer	MCU/ Silicon Core Number	Num- ber of Bits	Architecture	Most Notable Feature	Compatibility	IDEs	MRP (\$/€)	Use Case	Supported by
Raspberry Pi Compute Module 4 (CM4002008)	BCM2xxx	Raspberry Pi	Broad- com	BCM2711	64	ARM Cortex-A72	Inclusion of U.FL Antenna Connector	Connection with Arduino is possible	-	\$25 onwards	Applications in- volving gaming or networking	Raspberry Pi Com- munity
Portenta H7 (ABX00042)	STM32	Arduino	STMicro- electron- ics	STM- 32H747XI	32	ARM Cortex-M4F, Cortex-M7F	Two parallel cores that via a 'Remote Procedure Call' mechanism	Arduino IoT Cloud compatible	Arduino IDE (Desk- top), Ar- duino IDE (Cloud)	\$103	Parallel computing and multitasking operations like robotics controllers	Arduino Forum
ZM1-DB	ESP32	ZERYNTH	-	ESP32- D0WD	32	-	Extremely cus- tomisable due to the presence of zBus. It also has a crypto IC ATECC608A from Microchip	Connections available with a vast variety of expansion boards	Zerynth SDK (VSCode extension is avail- able)	\$127	IIoT ap- plications that involve transfer of sensitive data	Zerynth Commu- nity and ESP Com- munity
NUCLEO- L4P5ZG	STM32 Nucleo- 144	STMicro- electron- ics	STMicro- electron- ics	STM32L4P- 5ZGT6U	32	ARM Cortex-M4	Low power consumption	Supports Arduino, ST Zio and morpho connectivity	IAR Embedded Work- bench, MDK-ARM, and STM- 32CubeIDE	\$19	Applications with I/O management (when there are lots of sensors, actuators and peripherals)	ST Com- munity
FRDM-KE15Z	Kinetis-E	NXP	NXP	MKE15Z- 256VLL7	32	ARM Cortex- M0+	Has a complete set of ana- logue/digital features	Compatible with FRDM- TOUCH, FRDM-MC- LVBLDC and Arduino boards	IAR Embedded Workbench	\$50	Embedded systems de- velopment and prototyping	NXP Com- munity
BBONE-AI	AM57x	BEAGLE- BOARD	Texas Instru- ments	AM5729	32	ARM Cortex-A15	It is possible to run a neural network	Has dual C66 DSP, four PRU and four Embedded Vision engines	-	\$127	AI, ML and time-critical tasks like motor drivers	Beagle- Board Forum
SIMBA-DKL	-	Sen- siEDGE	STMicro- electron- ics	STM32L- 476RG	32	ARM Cortex-M4	Has a variety of sensors and peripherals on board	Its develop- ment kit, SIMBA-DKL comes with Arduino compatible connectors	Arduino IDE is compatible with its dev kit	\$106	Can be used as a sensor node in IoT based data acquisition systems	Partners with many commu- nity forms including Arrow. com
KITXM- C47RELAXV- 1TOB01	XMC4000	Infineon Technolo- gies	Infineon Technolo- gies	XMC4700- F144	32	ARM Cortex-M4	MicroSD card slot makes firmware up- dates easier	Arduino hardware compatible	DAVE IDE by Infineon	€30	IIoT applica- tions, can be used as a node	Infineon Developer Commu- nity
Nvidia AGX Xavier Industrial	-	Nvidia	Nvidia	8-core NVIDIA Carmel Armv8.2 64-bit CPU	64	Armv8.2	Contains AI software that can be used 'on the edge'	-	-	\$1,449	AI-powered applications like robots, drones and autonomous machines	Jetson Commu- nity
Particle Argon	-	Particle IO	-	Espressif ESP32- D0WD, Nordic Semi- conductor nRF52840 SoC	32	ARM Cortex-M4F	Allows the creation of MESH net- works	Supports parts of the Arduino API	Particle Web IDE, Particle Workbench	\$28	Battery operated Mesh networks for IoT applications	Particle Commu- nity